

PIGSTI

1 Preprocessing

Raw reads

(FASTQ)

Trim adapters

(Adapter Removal)

2 Host route

Host identification

(FastQ Screen)

Map host

(BWA / Bowtie2)

Map mtDNA

(BWA / Bowtie2)

Merge samples

(samtools)

Soft clipping

(Python)

Damage

(DamageProfiler)

Host species



Pathogen detection

(E-value · MaltExtract)



Pathogen mapping

(BWA / Bowtie2)



Authentication

(composite score)

Authentication details

ANI

Entropy

Breadth

5' C→T damage

Edit distance

Mapping ratio

Genus rank

Verifiable

(Excel · PDF)

3 Metagenomics screening

Dedup

(PRINSEQ++)

Pool samples

(cat · pigz)

Metagenome screening

(KrakenUniq)

Mammals removal

(Bowtie2)

Metagenome screening

(MALT)

Genetic sexing

(sexing_residual)

Preprocessing

Metagenomics

Mapping

Optional

Host

Detection

Authentication